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SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: B.E. / B.Tech
Course Code: **MLA02**
Course Name: **FUNDAMENTALS OF MACHINE LEARNING**
Duration: **1:30 Hr**
Date : _____ Max Marks: **50 Marks**

ASSIGNMENT – II

UNIT II – SUPERVISED LEARNING

Topics: *Linear regression and classification, naive Bayes, discriminant functions, logistic regression, decision trees and pruning, neural networks and back-propagation, SVM, ensembles (bagging, boosting)*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO2	Apply supervised learning algorithms for regression, classification, and predictive modeling tasks to solve structured data problems in practical applications.

Answer all the Tasks

Task 1. The following data relate the number of training epochs x to a validation score y.

x	1	2	3	4	5
y	2.1	3.9	6.2	7.8	10.1

(a) Fit $y = w_0 + w_1x$ by ordinary least squares using the normal equations. (b) Predict y at $x = 6$. (c) Compute the residual sum of squares, MSE and R^2. (d) Analyse the reliability of predicting at $x = 6$.

(5) (CO2) (BL4)

Task 2. For mean-centred data with $S_{xx} = \Sigma x^2 = 10$ and $S_{xy} = \Sigma xy = 19.9$ (single feature, no intercept needed), compute the ridge-regression slope for $\lambda = 5$. Calculate the percentage shrinkage relative to the least-squares slope and explain the bias–variance effect of increasing λ.

(2) (CO2) (BL4)

Task 3. An e-mail filter uses two binary features: Offer (contains the word 'offer') and Link (contains a hyperlink). Ten labelled e-mails are available.

#	Offer	Link	Class
1	Y	Y	Spam
2	Y	N	Spam
3	Y	Y	Spam
4	N	Y	Spam
5	Y	Y	Spam

#	Offer	Link	Class
6	N	N	Ham
7	N	Y	Ham
8	Y	N	Ham
9	N	N	Ham
10	N	N	Ham

(a) Estimate the priors and the class-conditional probabilities without smoothing. (b) Classify an e-mail with Offer = Y, Link = Y and report the posterior P(Spam | x). (c) Repeat (b) with Laplace smoothing (α = 1). (d) Classify an e-mail with Offer = N, Link = N (with smoothing) and analyse the effect of the conditional-independence assumption and of smoothing.

(6) (CO2) (BL4)

Task 4. A logistic-regression classifier $p = \sigma(w_1x_1 + w_2x_2 + b)$ is initialised with $w = (0, 0)$, $b = 0$ and trained on two samples: $x^{(1)} = (1, 2)$ with $y = 1$, and $x^{(2)} = (2, 1)$ with $y = 0$. Perform one batch gradient-descent step with learning rate $\eta = 0.5$ on the mean cross-entropy loss. Report the gradient, the updated parameters, the predicted probabilities and the loss before and after the update, and analyse the result.

(4) (CO2) (BL4)

Task 5. A node contains 10 training samples (6 Yes, 4 No). A candidate attribute A splits it into $A = a_1$ (4 samples: 4 Yes, 0 No) and $A = a_2$ (6 samples: 2 Yes, 4 No). Calculate the entropy of the node, the information gain of the split and the Gini-impurity decrease, and evaluate whether the split is worthwhile.

(3) (CO2) (BL4)

Task 6. In a decision tree trained on 100 samples, an internal node t with a four-leaf subtree T_t makes 3 training errors, whereas collapsing t into a single leaf would make 8 errors. Using cost-complexity pruning with $R_\alpha(T) = R(T) + \alpha|\text{leaves}|$, compute the critical value α at which the subtree and the single leaf have equal cost, and evaluate whether the subtree should be pruned for $\alpha = 0.01$ and $\alpha = 0.02$.

(3) (CO2) (BL5)

Task 7. A network has two inputs, one sigmoid hidden unit and one sigmoid output unit (no biases): $h = \sigma(w_1x_1 + w_2x_2)$, $\hat{y} = \sigma(vh)$, with $w_1 = 0.4$, $w_2 = 0.3$, $v = 0.5$. For the input $x = (1, 2)$ with target $t = 1$ and squared-error loss $E = \frac{1}{2}(t - \hat{y})^2$, perform one forward pass and one back-propagation update with $\eta = 0.5$. Report h, $\hat{y}$, E, the error terms δ, the gradients, the updated weights and the new loss.

(4) (CO2) (BL4)

Task 8. A linear SVM in two dimensions has $w = (0.5, 0.5)$ and $b = -2$, with the support vectors (3, 3) (class +1) and (1, 1) (class −1). (a) Verify that the support vectors lie on the margin hyperplanes and compute the geometric margin width. (b) Classify the points (4, 2) and (1, 2) and compute their distances from the decision boundary. (c) Compute the hinge loss for (1, 2) with true label −1 and for (2, 2) with true label +1, and state which points violate the margin.

(4) (CO2) (BL4)

Task 9. Independent classifiers, each with accuracy 0.70, are combined by majority voting. Calculate the ensemble accuracy for 3 and 5 classifiers, comment on the gain from adding more classifiers, and analyse what happens to this gain if the classifiers' errors are strongly correlated.

(3) (CO2) (BL4)

Task 10. AdaBoost starts with five training samples of equal weight. The first weak learner misclassifies samples 2 and 5. Compute the weighted error, the learner weight α, the un-normalised and normalised weights after the update, and analyse how boosting changes the focus of the next learner.

(4) (CO2) (BL4)

Task 11. Two classes share the covariance matrix $\Sigma = [[2, 0.5],[0.5, 1]]$ and have means $\mu_1 = (1, 2)^T$ and $\mu_2 = (3, 1)^T$. (a) Derive the discriminant direction $w = \Sigma^{-1}(\mu_2 - \mu_1)$ and the decision threshold for equal priors. (b) Classify $x = (2.5, 1)$. (c) Evaluate whether the decision changes if the priors are $P(\omega_1) = 0.8$ and $P(\omega_2) = 0.2$.

(4) (CO2) (BL4)

Task 12. A bank evaluates three classifiers on 1,000 test loans (100 actual defaulters). Default is the positive class.

Model	TP	FN	FP	TN
A	70	30	90	810
B	85	15	180	720
C	60	40	30	870

A missed defaulter (FN) costs ₹50,000 and a wrongly rejected good applicant (FP) costs ₹5,000.

(a) Compute accuracy, precision, recall and F1-score for each model. [3]

(b) Compute the total expected cost of each model. [2]

(c) Critically evaluate which model should be deployed and explain why accuracy and F1 lead to a different choice than cost. [2]

(d) Propose two modelling or threshold-tuning steps to improve the chosen model and one validation safeguard against over-fitting. [1]

(8) (CO2) (BL6)

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ANSWER SCRIPT EVALUATION SHEET
ASSIGNMENT - II

Programme	B.E / B.Tech	Course Code	MLA02
Course Name	Fundamentals of Machine learning.	Date	
Duration	1 Hr 30 Min	Max Marks	50

Register No.: 1911260190

TASK 1	Max Marks	Marks Awarded
Task 1	5	5
Task 2	2	2
Task 3	6	6
Task 4	4	4
Task 5	3	3
Task 6	3	3
Task 7	4	4
Task 8	4	4
Task 9	3	3
Task 10	4	2½
Task 11	4	3½
Task 12	8	7½
TOTAL	50	48½

Total in words:

Assignment -2

Unit -2

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given:

x	1	2	3	4	5
y	2.1	3.9	6.2	7.8	10.1

The linear regression

$$y = \omega_0 + \omega_1 x$$

5

$$\omega = (X^T X)^{-1} X^T y$$

$\Sigma x = 15$, $\Sigma y = 30.1$

$\Sigma x^2 = 55$, $\Sigma xy = 110.9$

$$5\omega_0 + 15\omega_1 = 30.1$$

$$15\omega_0 + 55\omega_1 = 110.9$$

$$\omega_0 = 0.05, \omega_1 = 1.99$$

$$y = 0.05 + 1.99x$$

b. at $x = 6$

$$y = 0.05 + 1.99(6)$$

$$= 0.05 + 11.94$$

$$y = 11.99$$

∴ Residual sum of squares:

$$RSS = \sum (y_i - \hat{y}_i)^2$$

∴ Predicted value of

$$RSS = 0.107$$

$$MSE = \frac{RSS}{n} = \frac{0.107}{5} = 0.0214$$

$$R^2 = 1 - \frac{RSS}{TSS} = 0.9973$$

Conclusion

The model provides a good fit, but the prediction at $x = 6$

$S_{xx} = 10$, $S_{xy} = 19.9$, $\lambda = 5$

$$w_1 = \frac{S_{xy}}{S_{xx} + \lambda} = \frac{19.9}{10+5} = 1.3267$$

2 $w_{OLS} = \frac{19.9}{10} = 1.99$

% of shrinkage:

$$\frac{1.99 - 1.3267}{1.99} \times 100 = 33.3\%$$

Bias - variance effect

Increasing λ reduces the magnitude of the regression coefficient

It increase bias but decrease variance

It helps prevent overfitting and improves

3. given:

There are 10 labelled emails

- Spam = 5
- Ham = 5

$P(\text{spam}) = 0.5$

$P(\text{Ham}) = 0.5$

Conditional probabilities	Spam	Ham
$P(\text{offer} = Y \mid \text{spam})$	0.8	0.2
$P(\text{offer} = N)$	0.2	0.8
$P(\text{link} = Y)$	0.8	0.2
$P(\text{link} = N)$	0.2	0.8

b. $P(\text{spam} \mid X) \propto P(\text{spam}) P(Y \mid \text{spam}) P(Y \mid \text{spam})$

$= 0.5 \times 0.8 \times 0.8 = 0.32$

for ham:

$= 0.5 \times 0.2 \times 0.2 = 0.02$

normalising:

$$P(\text{Spam}/X) = \frac{0.32}{0.32 + 0.02} = 0.9412$$

$$P = \frac{\text{Count} + \alpha}{N + \alpha k}$$

For Spam

$k = 2$

$$P(\text{Offer} = Y / \text{Spam}) = \frac{4+1}{5+2} = \frac{5}{7}$$

For Ham

$$P(\text{Offer} = Y / \text{Ham}) = \frac{1+1}{5+2} = \frac{2}{7}$$

$$P(\text{Link} = Y / \text{Ham}) = \frac{2}{7}$$

Spam Score:

$$0.5 \times \frac{5}{5} \times \frac{5}{7} = \frac{25}{98}$$

Ham Score

$$0.5 \times \frac{2}{7} \times \frac{2}{7} \times \frac{4}{98}$$

$$P(\text{Spam}/X) = \frac{25}{29} = 0.8621$$

3. d. for Spam:

$$P(N/\text{spam}) = \frac{2}{7}$$

Spam Score:

$$0.5 \times \frac{2}{7} \times \frac{2}{7} = \frac{4}{98}$$

For Ham

$$P(N/\text{Ham}) = \frac{5}{7}$$

Ham Score

$$0.5 \times \frac{5}{7} \times \frac{5}{7} = \frac{25}{98}$$

$$\therefore P(\text{Spam}/X) = \frac{4}{29} = 0.1379$$

$$P(\text{Ham}/X) = 0.8621$$

Conclusion:

The email is classified as Ham because it contains neither the offensive word nor a hyperlink.

given

$w_1 = 0$, $w_2 = 0$, $b = 0$

$\eta = 0.5$

$x^1 = (1, 2)$, $y^{(1)} = 1$

$x^2 = (2, 1)$, $y^2 = 0$

Sigmoid function:

$$a(z) = \frac{1}{1+e^{-z}}$$

$z = 0$

$P_1 = P_2 = 0.5$

$$\frac{\partial L}{\partial w} = \frac{1}{n} \Sigma (P - y) x$$

w_1:

$$\frac{(-0.5)(1) + (0.5)(2)}{2}$$

$$\frac{\partial L}{\partial w_1} = 0.25$$

w_2 :

$$= \frac{-0.5 \times 2 + 0.5 \times 1}{2}$$

$$\frac{dL}{dw_2} = -0.25$$

for bias:

$$\frac{dL}{db} = 0$$

$$w_{new} = w_{old} - \eta \Delta L$$

$$w_1 = 0 - 0.5(0.25) = -0.125$$

$$w_2 = 0.125$$

$$b = 0$$

$$w = (-0.125, 0.125), b = 0$$

$$z_1 = 0.125$$

$$P_1 = \sigma(0.125)$$

$$P_1 = 0.5312$$

$z_1 = -0.12$

$p_1 = 0.4684$

$L = -\ln(0.5) = 0.693$

London ✓

gradient descent successfully updates the parameter and reduces classification error.

5 given

Total = 10 (6Y, 4N)

$H(S) = -0.6 \log_2 0.6 - 0.4 \log_2 0.4$

$H(S) = 0.971$ ✓

IG:

$A_1 = 0$

A_2 :

$H(A_2) = -\frac{2}{2} \log_2 \frac{1}{6} - \frac{4}{6} \log_2 \frac{4}{4}$

$H(A_2) = 0.9183$

weighted entropy

$= 0.4 \times 0 + 0.6 \times 0.9183 = 0.551$

$IG = 0.420$

gini Impurity

$G(S) = 1 - (0.6)^2 - (0.4)^2 = 0.48$

∴ Gini Index $= 0.2133$

6. given:

Subtree error = 3, number of leaves = 4, Single leaf error = 8

Formula

$R_\alpha(T) = R(T) + \alpha(\text{leaves})$

$R_\alpha(T) = 3 + 4\alpha$

$R_\alpha(\text{leaf}) = 8 + \alpha$

$3 + 4\alpha = 8 + \alpha$

$\alpha = 1.667$

$\alpha = 0.01$

Subtree cost = 3.04

Single leaf = 8.01

Not pure

Given:

$w = [0.5, 0.5]$, $b = -2$

$f(x) = w^T x + b$

for (3,3):

$f(x) = 0.5(3) + 0.5(3) - 2 = 1$

for (1,1):

$f(x) = 0.5(1) + 0.5(1) - 2 = -1$

$$d = \frac{|w^T x + b|}{\|w\|}$$

$f(x) = 0.5(4) + 0.5(2) - 2 = 1$

class = +1

$d = 1.4142$

for (1,2):

$f(x) = 0.5(1) + 0.5(2) - 2 = -0.5$

class = -1

$d = 0.7071$

$$w_{new} = w_{old} - \frac{n dE}{dw}$$

$w_1 = 0.40487$

$w_2 = 0.30974$

$v = 0.53621$

$\hat{y}' = 0.59738$

$E' = \frac{1}{2}(1 - 0.59738)^2$

$E' = 0.08105$

Given:

$x = (1, 2)$, $t = 1$

$w_1 = 0.4$, $w_2 = 0.3$, $b = 0.5$

$LR = 0.5$

$$a(2) = \frac{1}{1+e^{-2}}$$

$$h = a(0.4 + 0.6) = 0.7311$$

$$y^{\wedge} = a(0.5 \times 0.7311)$$

$$y^{\wedge} = 0.5904$$

$$E = \frac{1}{2}(t - y^{\wedge})^2$$

$$= 0.0839$$

$$\delta_0 = (t - y^{\wedge}) y^{\wedge} (1 - y^{\wedge})$$

$$= 0.09906$$

$$\delta_h = h(1-h) v \delta_0$$

$$\delta_h = 0.009738$$

11. $\Sigma = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}$

$\mu_1 = (1, 2)$, $\mu_2 = (3, 1)$

a) $\Sigma^{-1} = \frac{1}{1.75} \begin{bmatrix} 1 & -0.5 \\ -0.5 & 2 \end{bmatrix}$

$\mu_2 - \mu_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$

$w = \Sigma^{-1}(\mu_2 - \mu_1)$

$w = (1.4286 \quad -1.7143)$

$c = \frac{1}{2}(\mu_2^T \Sigma^{-1} \mu_2 - \mu_1^T \Sigma^{-1} \mu_1)$

$= 0.2857$

$x = (2.5, 1)$

$w^T x = 1.4286(2.5) - 1.7143(1)$

$= 1.8571$

$1.8571 > 0.2857$

$\therefore$ point belong to class ω_2

$w_i = \frac{1}{5} = 0.2$

$\epsilon = 0.2 + 0.2 = 0.4$

$\alpha = \frac{1}{2} \ln\left(\frac{1-\epsilon}{\epsilon}\right)$

$= \frac{1}{2} \ln \frac{0.6}{0.4}$

$\alpha = 0.2027$

$0.2 e^{0.2027} = 0.24495$

$0.2 e^{-0.2027} = 0.16330$

c. Hinge loss:

$$L = \max(0, 1 - y f(x))$$

Err $L = 0.5$

for $(2, 2)$ $y = +1$

$$f(x) = 0$$

$$L = \max(0, 1) = 1$$

9. given:

$$p = 0.70, \quad q = 0.30$$

$$P(3) = \binom{3}{2} (0.7)^2 (0.3)^{1} (0.7)^3$$

$$P(3) = 78.4\%$$

$$P(5) = \binom{5}{3} (0.7)^3 (0.3)^2 + \binom{5}{4} (0.7)^4 (0.3)^1 + (0.7)^5$$

$$= 83.692\%$$

$P(\omega_1) = 0.8$, $P(\omega_2) = 0.2$

$$C' = 0.2917 + \ln \frac{0.8}{0.2}$$

$$C' = 1.6720$$

∴ object belongs to ω_2

Formula :

$$\text{Accuracy} = \frac{TP + TN}{\text{Total}}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2PR}{P + R}$$

Metric	A	B	C
Accuracy	88%	80.5%	93%
Precision	43.75%	32.08%	66.67%
Recall	70%	85%	60%
F1-Score	53.85%	46.58%	63.13%

b. Cost = (FN × 50000) + (FP × 5000)

Model A:

= (30 × 50000) + (90 × 5000)

= ₹19, 50, 000

Model B

= 15 × 50000 + 180 × 500

= 16, 50, 000

Model C

= (60 × 50000) + 30 × 5000

= 21, 50, 000

c. Model B is selected because it has lowest expected cost and highest recall of 95%.

Model C has higher accuracy and F1-score but it financial cost is high accuracy and F1 do not consider the different cost of false positive and false negative

Apply class-weighted learning to give more importance

Adjust the classification threshold to minimize false cost

Conclusion:

Model B is suitable because it minimizes the bank's expected financial loss.